Optimal Control and Reinforcement Learning
Introduction to Optimal Control

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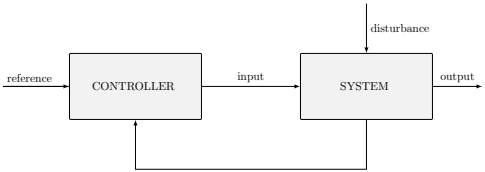
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Optimal Control and Reinforcement Learning @ University of Bologna.

Optimal Control

So far, feedback control design paradigm

- given a "control system"
  (model of an actual systems)
- goal: track a desired/reference output
  (or stabilize an equilibrium)
- design input as function of output/state



Two ingredients are mainly missing

- performance index
- constraints
  (input limitations, geometric constraints,...)

Optimization:
theory to find "best" decision while handling constraints

pervasive in Engineering... and several other fields!

Optimal Control: combine optimization with (feedback) control!

Optimal Control: motivating domains

Aircraft systems

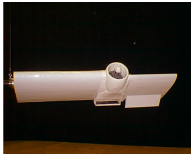
Performance index

- distance from desired "behavior" (position path, velocities, ...)
- minimum time
- fuel/battery consumption
- passenger confort

Constraints

- geometric (obstacles, attitude, stall limits,...)
- inputs (thrust, flaps)
- internal systems (angular and linear velocities, aerodynamic limits,...)





Optimal Control: motivating domains

Robotic systems

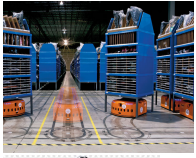
Performance index

- distance from desired "behavior"
  (evolution of end-effector, position path,...)
- minimum time
- ...

Constraints

- geometric (workspace, obstacles, payload geometry, ...)
- inputs (motor saturations)
- internal systems (joint constraints, velocities,...)







Optimal Control: motivating domains

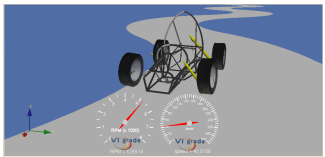
Autonomous driving

Performance index

- distance from desired "behavior" (lane center, velocity,...)
- minimum time (race vehicles)
- passengers confort
- fuel consumption

Constraints

- geometric (lane, obstacles,...)
- inputs (throttle, steering wheel)
- internal systems (tire limits, normal load, velocities,...)



Optimal Control: motivating domains

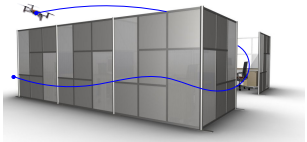
Autonomous aerial vehicles (drones)

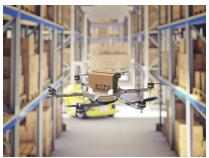
Performance index

- distance from desired "behavior" (given 3D path, velocities,...)
- minimum time
- fuel/battery consumption
- exploration cost

Constraints

- geometric (buildings, other obstacles, attitude,...)
- inputs (thrust)
- internal systems (angular and linear velocities,...)





Optimal Control: motivating domains

Space systems

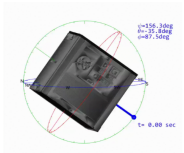
Performance index

- distance from desired "behavior" (desired orbit, velocities,...)
- energy consumption
- minimum time

Constraints

- geometric (orbit, orientation wrt sun/Earth, ...)
- inputs (thrust)
- internal systems (angular and linear velocities,...)





Continuous-time control systems

Consider the continuous-time, $t \in \mathbb{R}$, (time-varying) system

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t), t), \qquad x(0) = x^0 \qquad \text{state equation} \\ y(t) &= h(x(t), u(t), t) \qquad\qquad\qquad\quad \text{output equation} \end{aligned}$$

- $x(t) \in \mathbb{R}^n$ *state* of the system at time t
- $u(t) \in \mathbb{R}^m$ *input* of the system at time t
- $y(t) \in \mathbb{R}^p$ *output* of the system at time t

$$x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix} \qquad u(t) = \begin{bmatrix} u_1(t) \\ \vdots \\ u_m(t) \end{bmatrix} \qquad y(t) = \begin{bmatrix} y_1(t) \\ \vdots \\ y_p(t) \end{bmatrix}$$

Remark In applications, we will mainly work with time-invariant systems, $\dot{x}(t) = f(x(t), u(t))$, even though time-varying ones may be considered. □

Discrete-time control systems and notation

We consider nonlinear, discrete-time systems described by

$$x(t+1) = f(x(t), u(t), t), \qquad x(0) = x^0, \qquad t \in \mathbb{N}_0$$

but from now on we will use the (more compact) notation

$$x_{t+1} = f_t(x_t, u_t), \qquad x_0 = x^0, \qquad t \in \mathbb{N}_0$$

where $x_t \in \mathbb{R}^n$ and $u_t \in \mathbb{R}^m$ are the state and the input of the system at time t.

Discrete-time control systems and notation

Consider a nonlinear, discrete-time system on a finite time-horizon

$$x_{t+1} = f_t(x_t, u_t), \qquad x_0 = x^0, \qquad t = 0, \dots, T-1 \tag{1}$$

where $x_t \in \mathbb{R}^n$ and $u_t \in \mathbb{R}^m$ are the state and the input of the system at time t.

Notation

We use $\mathbf{x} \in \mathbb{R}^{nT}$ and $\mathbf{u} \in \mathbb{R}^{mT}$ to denote, respectively, the stack of the states x_t for all $t = 1, \dots, T$ and the inputs u_t for all $t = 0, \dots, T-1$, that is

$$\mathbf{x} := \mathrm{col}(x_1, \dots, x_T)$$
$$\mathbf{u} := \mathrm{col}(u_0, \dots, u_{T-1}).$$

where $\mathrm{col}(x_1, \dots, x_T)$ is the vector with components $x_1, \dots, x_T$.

System trajectories

Let us introduce the definition of trajectory of a (possibly) nonlinear, discrete-time control system.

Definition A pair $(\bar{\mathbf{x}}, \bar{\mathbf{u}}) \in \mathbb{R}^{nT} \times \mathbb{R}^{mT}$ is called a *trajectory* of system (1) if $\bar{x}_{t+1} = f_t(\bar{x}_t, \bar{u}_t)$ for all $t = 0, \dots, T-1$. In particular, $\bar{\mathbf{x}}$ is the state trajectory, while $\bar{\mathbf{u}}$ is the input trajectory. □

Let us recall a special class of trajectories characterized by constant input and constant state.

Definition A state-input pair $(x_e, u_e) \in \mathbb{R}^n \times \mathbb{R}^m$ is called an equilibrium pair of (1) if $(x_t, u_t) = (x_e, u_e)$, $t \in \mathbb{N}_0$, is a trajectory of the system. □

Equilibria of discerete-time, time-invariant systems satisfy $x_e = f(x_e, u_e)$.

Linearization of a system about a trajectory

Given the dynamics

$$x_{t+1} = f_t(x_t, u_t) \qquad t \in \mathbb{N}_0,$$

and a trajectory $(\bar{\mathbf{x}}, \bar{\mathbf{u}})$ (i.e., $\bar{x}_{t+1} = f_t(\bar{x}_t, \bar{u}_t)$ for all $t \in \mathbb{N}_0$), the linearization of (1) about $(\bar{\mathbf{x}}, \bar{\mathbf{u}})$ is given by the linear (possibly) time-varying system

$$\Delta x_{t+1} = A_t \Delta x_t + B_t \Delta u_t \qquad t \in \mathbb{N}_0,$$

with A_t and B_t the Jacobians of f_t, with respect to state and input respectively, evaluated at $(\bar{\mathbf{x}}, \bar{\mathbf{u}})$.

Recall some tools for nonlinear systems analysis and control

- System Trajectories and equilibria
- Linearization
- Equilibrium (origin) stabilization (typically time-invariant systems)
- Output/trajectory tracking (typically time-varying systems)

The following issues are not handled explicitly

- Optimize a performance index (e.g., control effort, distance from a target point, time, ...)
- Explicitly handle constraints (e.g., saturations on the inputs, collision/obstacle avoidance, regions where the state has to be confined,...)

Optimization: main ingredients

The main ingredients of an optimization problem are

Decision vector: $z \in \mathbb{R}^d$

Cost function: $\ell(z) : \mathbb{R}^d \to \mathbb{R}$ cost associated to decision z

Constraints (constraint sets): for some given functions $h_i : \mathbb{R}^n \to \mathbb{R}$, $i = 1, \dots, m$, and $g_j : \mathbb{R}^n \to \mathbb{R}$, the decision vector $z \in \mathbb{R}^d$ needs to satisfy

$$\begin{aligned} h_i(z) &= 0 \qquad i = 1, \dots, m \\ g_j(z) &\leq 0 \qquad j = 1, \dots, r. \end{aligned}$$

Equivalently we can say that we require $z \in Z$ with

$$Z = \{z \in \mathbb{R}^n \mid h(z) = 0, g(z) \leq 0\},$$

where we compactly denoted $h(z) = \mathrm{col}(h_1(z), \dots, h_m(z))$ and $g(z) = \mathrm{col}(g_1(z), \dots, g_r(z))$.

Example of cost function: scalar set up

Scalar set up $d = 1$

Decision variable: $z \in \mathbb{R}^d = \mathbb{R}$

Cost function: $\ell(z) = \frac{1}{2}(z - \bar{z})^2$, with $\bar{z} \in \mathbb{R}$ a give point.

Example of cost function: vector set up

Decision variable: $z \in \mathbb{R}^d$

Cost function:

- Distance from a point: $\ell(z) = \frac{1}{2}\|z - \bar{z}\|^2 = \frac{1}{2}(z - \bar{z})^\top (z - \bar{z})$ with $\bar{z} \in \mathbb{R}^d$ a given point.
- Weighted distance: given a positive (semi-)definite (weight) matrix $W \in \mathbb{R}^{d \times d}$,

$$\ell(z) = \frac{1}{2}\|z - \bar{z}\|_W^2 = \frac{1}{2}(z - \bar{z})^\top W (z - \bar{z}),$$

with $\bar{z} \in \mathbb{R}^d$ a given point.

Consider $z \in \mathbb{R}^2$, $z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$, $W = \begin{bmatrix} w_1 & 0 \\ 0 & w_2 \end{bmatrix} > 0$ with $w_1 > 0, w_2 > 0$, the cost function is

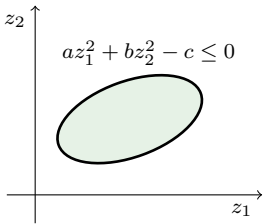
$$\ell(z) = \frac{1}{2}\|z - \bar{z}\|_W^2 = \frac{1}{2} w_1 (z_1 - \bar{z}_1)^2 + \frac{1}{2} w_1 (z_2 - \bar{z}_2)^2$$

Examples of constraint sets

Inequality constraint

$z \in \mathbb{R}^2$, such that $az_1^2 + bz_2^2 - c \leq 0$, for some given parameters $a \in \mathbb{R}, b \in \mathbb{R}, c \in \mathbb{R}$.
Then, the constraint set Z is

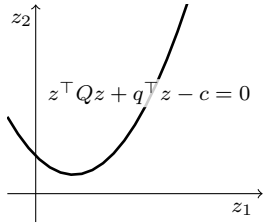
$$Z = \{z \in \mathbb{R}^2 \mid az_1^2 + bz_2^2 - c \leq 0\}.$$



Equality constraint

$z \in \mathbb{R}^d$, such that $z^\top Q z + q^\top z - c = 0$, for some given parameters $Q \in \mathbb{R}^{d \times d}, q \in \mathbb{R}^d, c \in \mathbb{R}$.
Then, the constraints set X is

$$Z = \{z \in \mathbb{R}^d \mid z^\top Q z + q^\top z - c = 0\}.$$



Optimization problem (minimization)

We can write our optimization problem as

$$\begin{aligned} \min_{z \in \mathbb{R}^d} \quad & \ell(z) \\ \text{subj.to} \quad & h_i(z) = 0 \qquad i = 1, \dots, m \\ & g_j(z) \leq 0 \qquad j = 1, \dots, r \end{aligned}$$

where $h_i : \mathbb{R}^d \to \mathbb{R}$ and $g_j : \mathbb{R}^d \to \mathbb{R}$.

We can write it more compactly as

$$\begin{aligned} \min_{z \in \mathbb{R}^d} \quad & \ell(z) \\ \text{subj.to} \quad & h(z) = 0 \\ & g(z) \leq 0 \end{aligned}$$

where $h : \mathbb{R}^d \to \mathbb{R}^m$ and $g : \mathbb{R}^d \to \mathbb{R}^r$.

Discrete-time optimal control: main ingredients (I)

The main ingredients of a discrete-time optimal control problem are

Dynamics: a (discrete-time) system in state space form

$$x_{t+1} = f_t(x_t, u_t) \qquad t = 0, \dots, T-1$$

We will mainly work on a finite horizon, i.e. $0 < T < +\infty$ (notation: $0, \dots, T-1$). In some cases we will also consider an infinite horizon, i.e., $T = +\infty$.

The dynamics introduces T equality constraints

$$\begin{aligned}
x_1 &= f(x_0, u_0) && \text{i.e.} && f_t(x_0, u_0) - x_1 = 0 \\
x_2 &= f(x_1, u_1) && \text{i.e.} && f_t(x_1, u_1) - x_2 = 0 \\
&\vdots && && \vdots \\
x_T &= f(x_{T-1}, u_{T-1}) && \text{i.e.} && f_t(x_{T-1}, u_{T-1}) - x_T = 0
\end{aligned}$$

Discrete-time optimal control: main ingredients (II)

Decision variable: state-input vector $(\mathbf{x}, \mathbf{u})$ and, if not given, initial state x_0.

Cost function: a cost "to be payed" for a chosen trajectory. We consider an additive structure in time

$$\ell(x_0, \mathbf{x}, \mathbf{u}) = \sum_{t=0}^{T-1} \ell_t(x_t, u_t) + \ell_T(x_T)$$

where $\ell_t : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is the so-called stage cost, while $\ell_T : \mathbb{R}^n \to \mathbb{R}$ is the terminal cost.

Notation: if the initial state is given, i.e., $x_0 = x^0$, we lighten the notation and write

$$\ell(\mathbf{x}, \mathbf{u}) = \sum_{t=0}^{T-1} \ell_t(x_t, u_t) + \ell_T(x_T)$$

Discrete-time optimal control: main ingredients (III)

End-point constraints: function of the state variable prescribed at initial and/or final point

$$r(x_0, x_T) = 0.$$

Path constraints: point-wise (in time) constraints representing possible limits on states and inputs at each time t

$$g_t(x_t, u_t) \leq 0, \quad t = 0, \ldots, T-1.$$

Cost function: examples

Distance from a desired state-input function of time ("state-input curve")

$$\ell(\mathbf{x},\mathbf{u}) = \sum_{t=0}^{T-1} \underbrace{\tfrac{1}{2}\|x_t - x_t^{\text{des}}\|_Q^2 + \tfrac{1}{2}\|u_t - u_t^{\text{des}}\|_R^2}_{\ell_t(x_t,u_t)} + \underbrace{\tfrac{1}{2}\|x_T - x_T^{\text{des}}\|_{Q_T}^2}_{\ell_T(x_T)}$$

where, $x_0 = x^0$, and $x_t^{\text{des}}, t = 1,\dots,T$, $u_t^{\text{des}}, t = 0,\dots,T-1$, (compactly $(\mathbf{x}^{\text{des}},\mathbf{u}^{\text{des}})$) represents a desired behavior for the system based on designers goals (not taking into account dynamic models).

Recall:

- $\frac{1}{2}\|x_t - x_t^{\text{des}}\|_Q^2 = \frac{1}{2}(x_t - x_t^{\text{des}})^\top Q(x_t - x_t^{\text{des}})$ for some Q positive (semi-)definite
- $\frac{1}{2}\|x_T - x_T^{\text{des}}\|_{Q_T}^2 = \frac{1}{2}(x_T - x_T^{\text{des}})^\top Q_T(x_T - x_T^{\text{des}})$ for some Q_T positive (semi-)definite
- $\frac{1}{2}\|u_t - u_t^{\text{des}}\|_R^2 = \frac{1}{2}(u_t - u_t^{\text{des}})^\top R(u_t - u_t^{\text{des}})$ for some R positive definite

Note: The desired curve $(\mathbf{x}^{\text{des}},\mathbf{u}^{\text{des}})$ does not satisfy the dynamics in general (e.g., a step transition between two equilibria).

Discrete-time optimal control problem

A discrete-time optimal control problem can be written as

$$\begin{aligned} \min_{\substack{x_0, x_1, \dots, x_T \\ u_0, \dots, u_{T-1}}} \quad & \sum_{t=0}^{T-1} \ell_t(x_t, u_t) + \ell_T(x_T) \\ \text{subj.to} \quad & x_{t+1} = f_t(x_t, u_t), \quad t = 0, \dots, T-1 \\ & r(x_0, x_T) = 0, \\ & g_t(x_t, u_t) \leq 0, \quad t = 0, \dots, T-1 \end{aligned}$$

where $\ell_t : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is the so-called stage cost while $\ell_T : \mathbb{R}^n \to \mathbb{R}$ is the terminal cost.

Trajectory generation

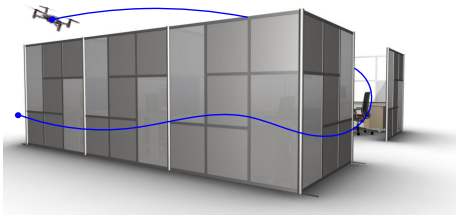
Consider the nonlinear, discrete-time system

$$x_{t+1} = f_t(x_t, u_t) \qquad t \in \mathbb{N}_0$$

and let

$$y_t = h_t(x_t, u_t) \qquad t \in \mathbb{N}_0$$

be a *performance output* of the system.



We would like the (performance) output to follow a desired time law y_t^{des}, for $t \in \mathbb{N}_0$.

Optimal Control Task

Find a (state-input) trajectory $(\mathbf{x}, \mathbf{u})$
i.e., a function of time (x_t, u_t), $t \in \mathbb{N}_0$, satisfying $x_{t+1} = f_t(x_t, u_t)$, for all $t \in \mathbb{N}_0$, such that

$$y_t \approx y_t^{\text{des}} \text{ for all } t \in \mathbb{N}_0.$$

Optimal control for trajectory generation

We can pose a trajectory generation problem as

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^{nT},\mathbf{u}\in\mathbb{R}^{mT}} \quad & \sum_{t=0}^{T-1} \frac{1}{2}||x_t - x_t^{\text{des}}||_Q^2 + \frac{1}{2}||u_t - u_t^{\text{des}}||_R^2 + \frac{1}{2}||x_T - x_T^{\text{des}}||_{Q_T}^2 \\ \text{subj.to} \quad & x_{t+1} = f_t(x_t, u_t), \quad t = 0, \dots, T-1 \\ & x_0 = x^0, \end{aligned}$$

with x^0 a given initial condition.

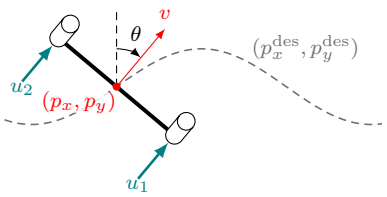
The state-input curve (function of time) $(x_t^{\text{des}}, u_t^{\text{des}})$, $t \in \mathbb{N}_0$, represents a desired behavior of the system and (in general) does NOT satisfy the dynamics.

The curve $(x_t^{\text{des}}, u_t^{\text{des}})$, $t \in \mathbb{N}_0$ is related to the performance output, i.e., $y_t^{\text{des}} = h_t(x_t^{\text{des}}, u_t^{\text{des}})$.

Trajectory generation for an autonomous drone

Consider a planar drone model (continuous-time to be discretized)

$$\ddot{p}_x = -\frac{u_1 + u_2}{M} \sin\theta$$
$$\ddot{p}_y = \frac{u_1 + u_2}{M} \cos\theta - g$$
$$\ddot{\theta} = \frac{L}{2J}(u_1 - u_2)$$



with performance output (discrete-time)

$$y_t = \begin{bmatrix} p_{x,t} \\ p_{y,t} \end{bmatrix} \qquad t \in \mathbb{N}_0$$

being, e.g., the position of the drone center of mass.

Goal:

We would like the position of the drone to follow a given time law in the 3D space possibly not exceeding some limits in the angles, in the angular velocities and in the inputs.

Example: trajectory generation for an autonomous drone (I)

Consider the discrete-time version of the drone dynamics (discretized via, e.g., forward Euler), i.e.

$$x_{t+1} = \underbrace{x_t + \delta f_{\text{CT}}(x_t, u_t, t)}_{f_t(x_t, u_t)}$$

where $\delta > 0$ sufficiently small discretization step and $f_{\text{CT}}(\cdot, \cdot, \cdot)$ continuous-time dynamics.

The state and input vector are:

$$x_t = \begin{bmatrix} p_{x,t} \\ v_{x,t} \\ p_{y,t} \\ v_{y,t} \\ \theta_t \\ \omega_t \end{bmatrix} = \begin{bmatrix} x_{1,t} \\ x_{2,t} \\ x_{3,t} \\ x_{4,t} \\ x_{5,t} \\ x_{6,t} \end{bmatrix} \qquad u_t = \begin{bmatrix} u_{1,t} \\ u_{2,t} \end{bmatrix}$$

Example: trajectory generation for an autonomous drone (II)

Trajectory generation task: compute a (state-input) trajectory of the drone such that

- the time-evolution of the center of mass is close to a desired one, $(p_{x,t}^{\text{des}}, p_{y,t}^{\text{des}})$ for all $t = 1, \dots, T$
- the velocity is close to the corresponding desired velocity profile $(v_{x,t}^{\text{des}}, v_{y,t}^{\text{des}})$ for all $t = 1, \dots, T$
- angle and angular velocity, θ_t, ω_t, are "not too high"
- inputs $u_{1,t}, u_{2,t}$ are "not too high".

Example: trajectory generation for an autonomous drone (III)

We can choose diagonal weighting matrices

$$Q = \begin{bmatrix} q_1 & & \\ & \ddots & \\ & & q_6 \end{bmatrix}, \; Q_T = \begin{bmatrix} q_{T1} & & \\ & \ddots & \\ & & q_{T6} \end{bmatrix}, \; R = \begin{bmatrix} r_1 & \\ & r_2 \end{bmatrix}.$$

The cost function then reads

$$\ell(\mathbf{x}, \mathbf{u}) = \sum_{t=0}^{T-1} \frac{1}{2} \Big[q_1 (x_{1,t} - p_{x,t}^{\text{des}})^2 + q_2 (x_{2,t} - v_{x,t}^{\text{des}})^2 + q_3 (x_{3,t} - p_{y,t}^{\text{des}})^2 + q_4 (x_{4,t} - v_{y,t}^{\text{des}})^2 + q_5 x_{5,t}^2 + q_6 x_{6,t}^2 + r_1 u_{1,t}^2 + r_2 u_{2,t}^2 \Big] + \frac{1}{2} \|x_T - x_T^{\text{des}}\|_{Q_T}^2$$

Example: path constraints

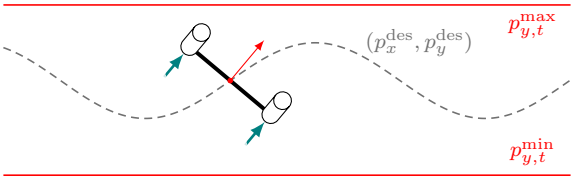
As for the (inequality) path constraints $g(\mathbf{x}, \mathbf{u}) \leq 0$, they represent restriction we want to (must) enforce at each time $t = 1, \dots, T$. For example:

- Saturation of the inputs

$$u_t^{\min} \leq u_t \leq u_t^{\max}$$

- State limitations, e.g. due to obstacles or limitation in the states. In the drone example, we may need to ensure

$$p_{y,t}^{\min} \leq p_{y,t} \leq p_{y,t}^{\max} \qquad (p_{y,t}^{\min} \leq x_{3,t} \leq p_{y,t}^{\max})$$



Continuous-time Optimal Control problem (I)

A continuous-time optimal control problem, i.e., $t \in \mathbb{R}$, can be written as

$$\begin{aligned} \min_{(x(\cdot),u(\cdot))\in\mathcal{F}} \quad & \int_0^T \ell_\tau(x(\tau),u(\tau))d\tau + \ell_T(x(T))) \\ \text{subj.to} \quad & \dot{x}(t) = f_t(x(t),u(t)) \qquad t \in [0,T] \\ & r(x(0),x(T)) = 0 \\ & g_t(x(t),u(t)) \leq 0 \qquad t \in [0,T) \end{aligned}$$

Note $\mathcal{F}$ is a space of functions (function space).

- Cost functional $\ell : \mathcal{F} \to \mathbb{R}$

$$\ell(x(\cdot),u(\cdot)) = \int_0^T \ell_\tau(x(\tau),u(\tau))d\tau + \ell_T(x(T))$$

- Space of trajectories (or *trajectory manifold*)

$$\mathcal{T} = \Big\{ (x(\cdot),u(\cdot))) \in \mathcal{F} \mid \dot{x}(t) = f_t(x(t),u(t)),\ t \in [0,T] \Big\}$$

Continuous-time Optimal Control problem (II)

Suppose we neglect path and terminal constraints. We can write the optimal control problem as

$$\begin{aligned} \min_{(x(\cdot),u(\cdot))\in\mathcal{F}} \quad & \ell(x(\cdot),u(\cdot)) \\ \text{subj.to} \quad & (x(\cdot),u(\cdot)) \in \mathcal{T} \end{aligned}$$

which is an infinite dimensional optimization problem.

Example: trajectory generation for a pendulum (I)

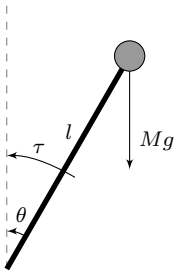
Consider a pendulum with (nonlinear) dynamics

$$\ddot{\theta} = -\frac{g}{l}\sin\theta + \frac{1}{m\ell^2}\tau$$

with state-space representation

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -\frac{g}{l}\sin x_1 + \frac{1}{m\ell^2}u$$

and performance output $y = \theta$.



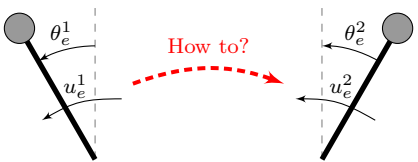
Example: trajectory generation for a pendulum (II)

Task

Transfer the pendulum from an equilibrium to another one.

Equilibrium conditions

$$0 = x_{1,e}$$
$$0 = -\frac{g}{l}\sin x_{1,e} + \frac{1}{m\ell^2}u_e$$



then, the first equilibrium is

$$x_e^1 = \begin{bmatrix}\theta_e^1 \\ 0\end{bmatrix} \qquad\qquad u_e^1 = mg\ell\sin\theta_e^1$$

A similar reasoning holds for the second equilibrium (x_e^2, u_e^2).

Example: trajectory generation for a pendulum (III)

In order to find a trajectory to move from an equilibrium to another one, we can define as desired curve

$$x_1^{\text{des}}(t) = \begin{cases} \theta_e^1 & 0 \le t < \frac{T}{2} \\ \theta_e^2 & \frac{T}{2} \le t \le T \end{cases}$$

$$x_2^{\text{des}}(t) = 0$$

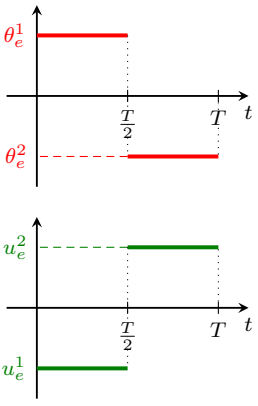
with desired input

$$u^{\text{des}}(t) = \begin{cases} u_e^1 & 0 \le t < \frac{T}{2} \\ u_e^2 & \frac{T}{2} \le t \le T \end{cases}$$

At the same time, we want $\dot{\theta}$ as small as possible.

IMPORTANT

$(x^{\text{des}}(t), u^{\text{des}}(t))$ is **not** a trajectory of the system!



Example: trajectory generation for a pendulum (IV)

The optimal control problem then reads

$$\begin{aligned}\min_{(x(\cdot),u(\cdot))\in\mathcal{F}} \quad & \int_0^T q_1(x_1(\tau)-x_1^{\text{des}}(\tau))^2 + q_2 x_2(\tau)^2 + r(u(\tau)-u^{\text{des}}(\tau))^2 d\tau \\ & + q_{T1}(x_1(T)-x_1^{\text{des}}(T))^2 + q_{T2}x_2(T)^2 \\ \text{subj.to} \quad & \begin{cases}\dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = -\frac{g}{l}\sin x_1(t) + \frac{1}{m\ell^2}u(t)\end{cases} \\ & x(0) = x^0\end{aligned}$$

where we chose as weight matrices $Q = \begin{bmatrix} q_1 & 0 \\ 0 & q_2 \end{bmatrix}$, $Q_T = \begin{bmatrix} q_{T1} & 0 \\ 0 & q_{T2} \end{bmatrix}$, $R = r$.